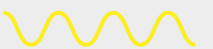
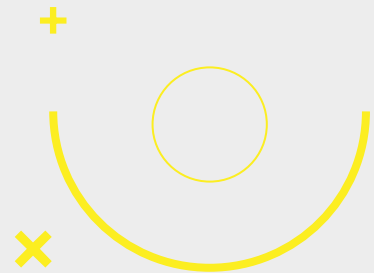




Aircraft Anti-icing Analysis: Water droplet dynamics under high frequency atomization and super hydrophobic effects

Kevin Thomas Fernandez

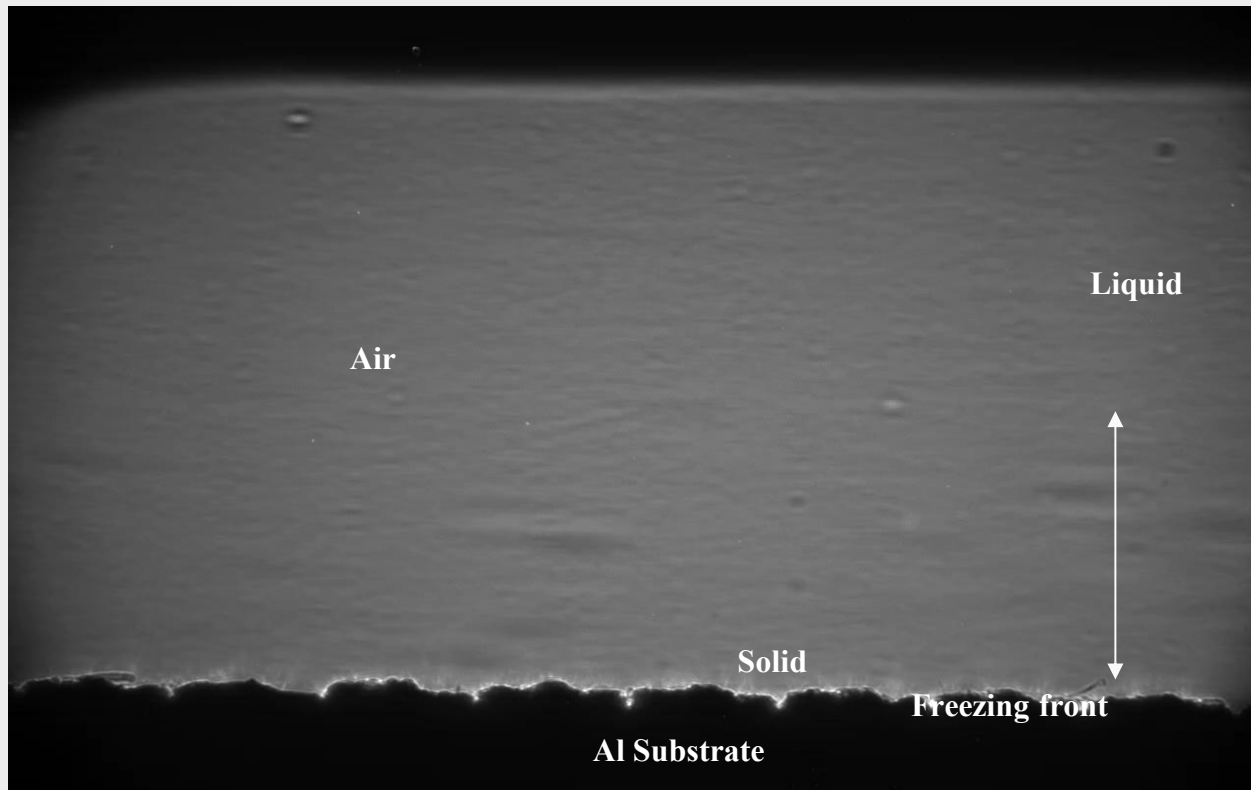
Graduate Research Symposium



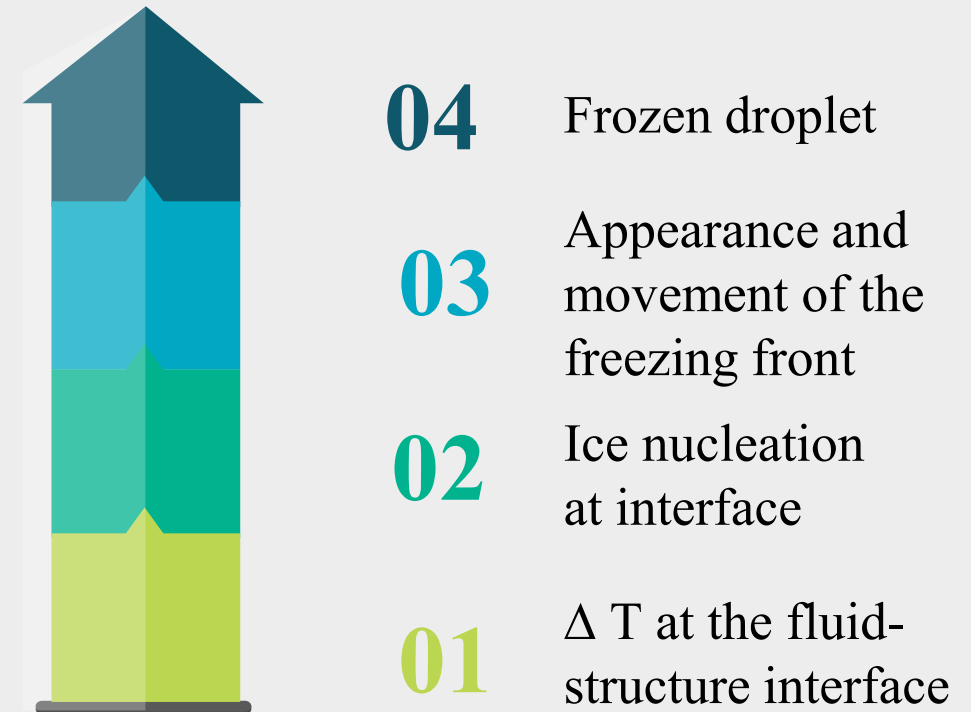
Overview

- **Motivation**
- **General Objectives**
- **Overview of approach**

Icing is the phenomenon of phase change from a liquid state to a solid state in sub-ambient conditions



4mm ambient water droplet, dropped at a height (Δd) of ~7 cm onto a Laser Etched Aluminum at -5°C



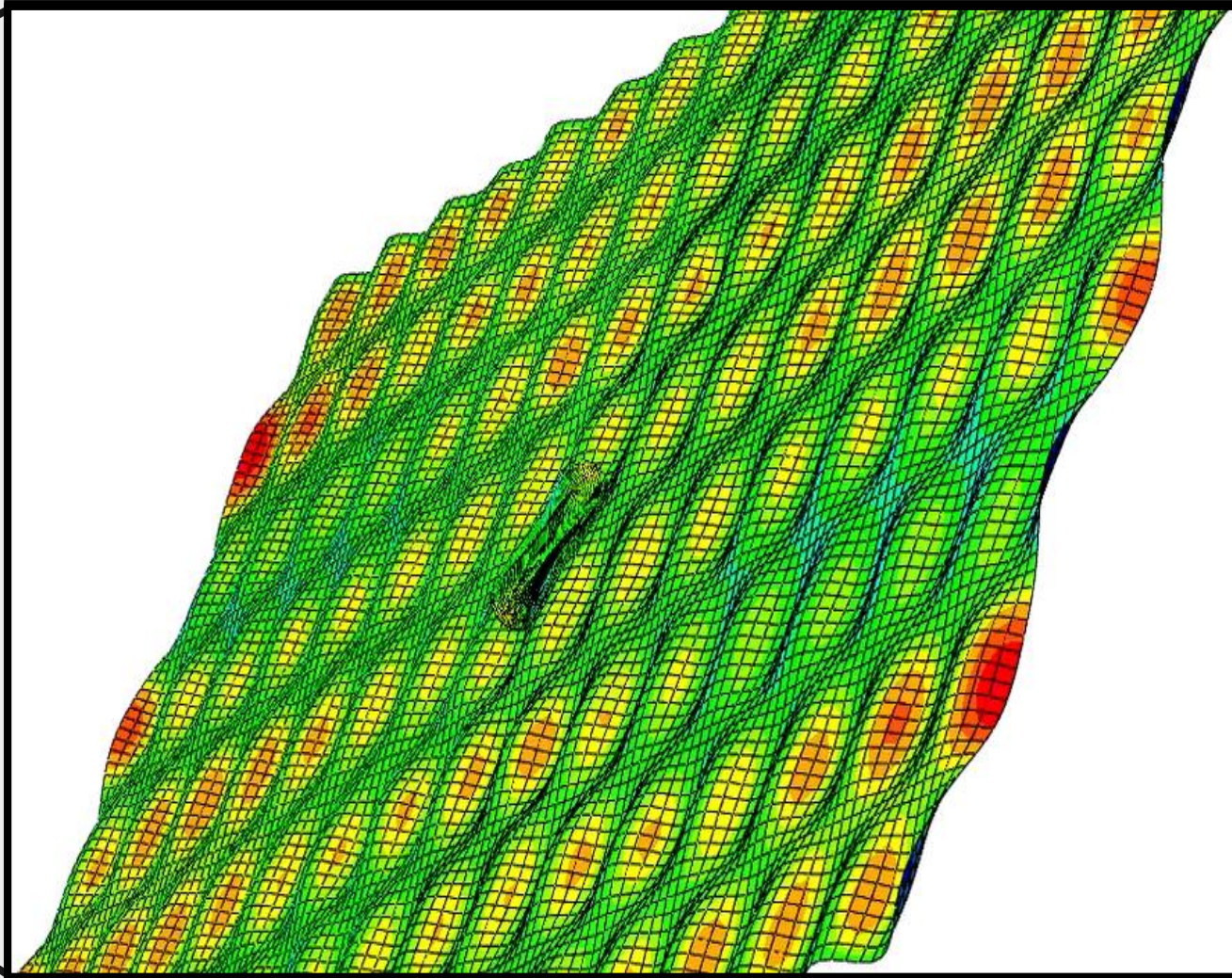
Challenges due to icing

- ☐ Structural damage
- ☐ Loss of control
- ☐ Increased drag and weight

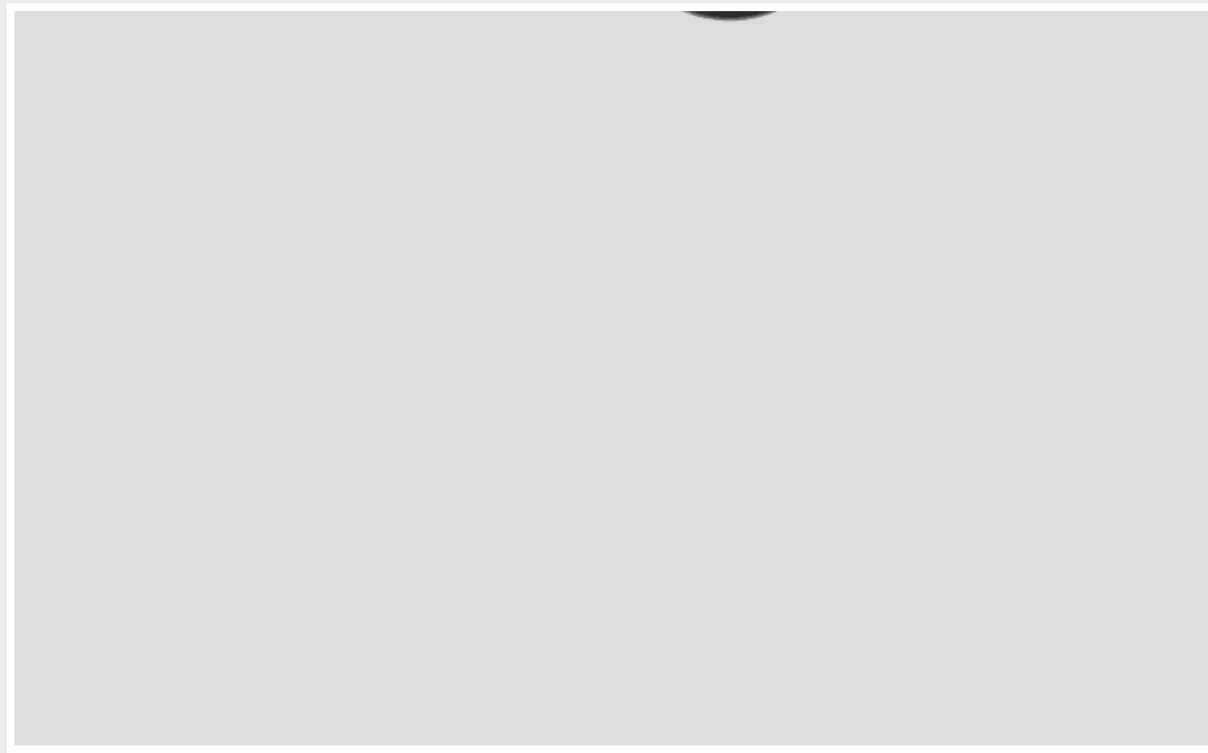
General Objectives

1. To introduce Atomization using Piezo Transducer (PZT) as an anti-icing approach
2. To study the effect of Atomization on the droplet impact characteristics at varied Re and setup protocols
3. To combine an active method and passive method

PZT Selection

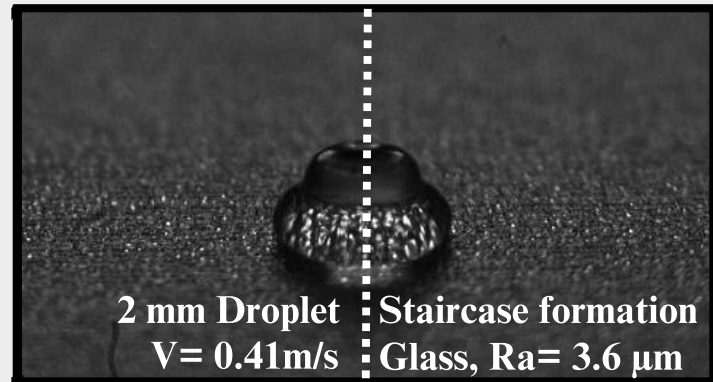


Atomization

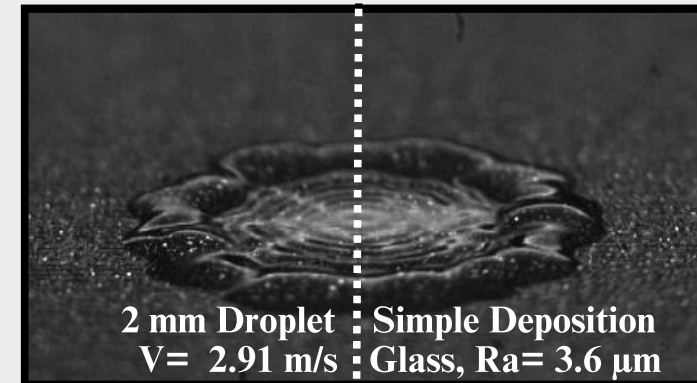


The breakup of a droplet at resonating frequencies into fine particles

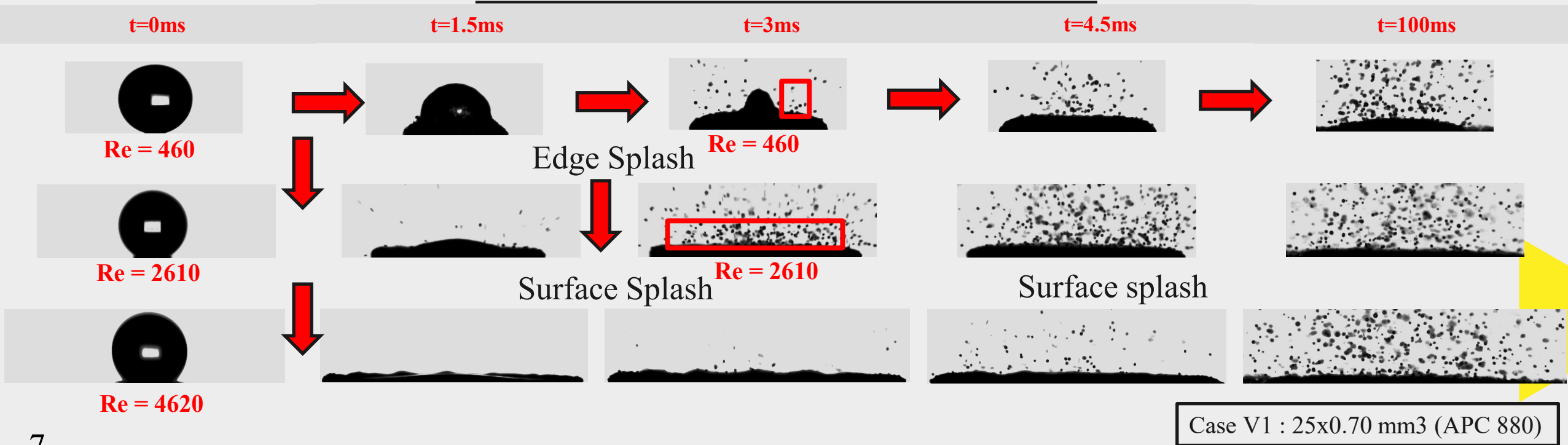
Droplet Impact: Variation in Re



$$Re = \frac{V_0 D_{droplet}}{\nu}$$



Case V1 Vibrated at 14128Hz



Single Frequency : Spread Diameter

Frequency : 14128Hz



Re= 460
Max Spread
 $D_{\max}=4.3$ mm

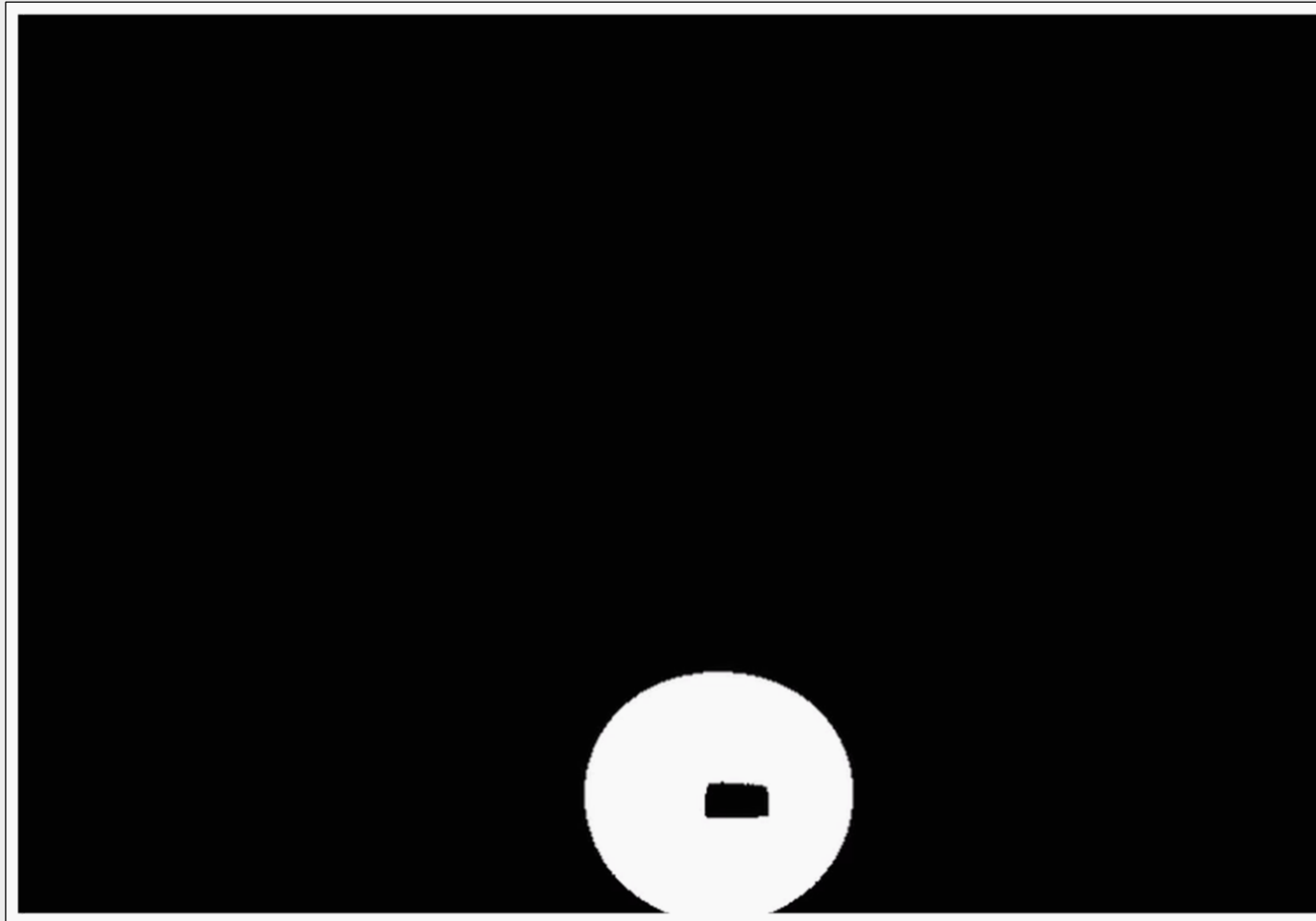


Re= 2610
Max Spread
 $D_{\max}=6.4$ mm



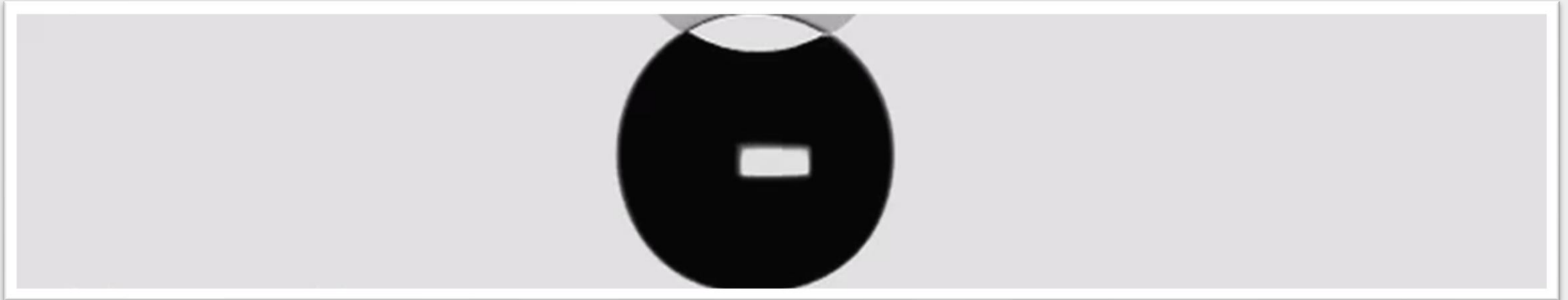
Re= 4610
Max Spread
 $D_{\max}=8.1$ mm

Single Frequency : Particle Tracking using ImageJ



Re= 460
Max Spread
 $D_{\max}$ =4.3 mm
PZT: 25mm Disc PZT,
Thickness: 0.7mm
Frequency : 14128Hz

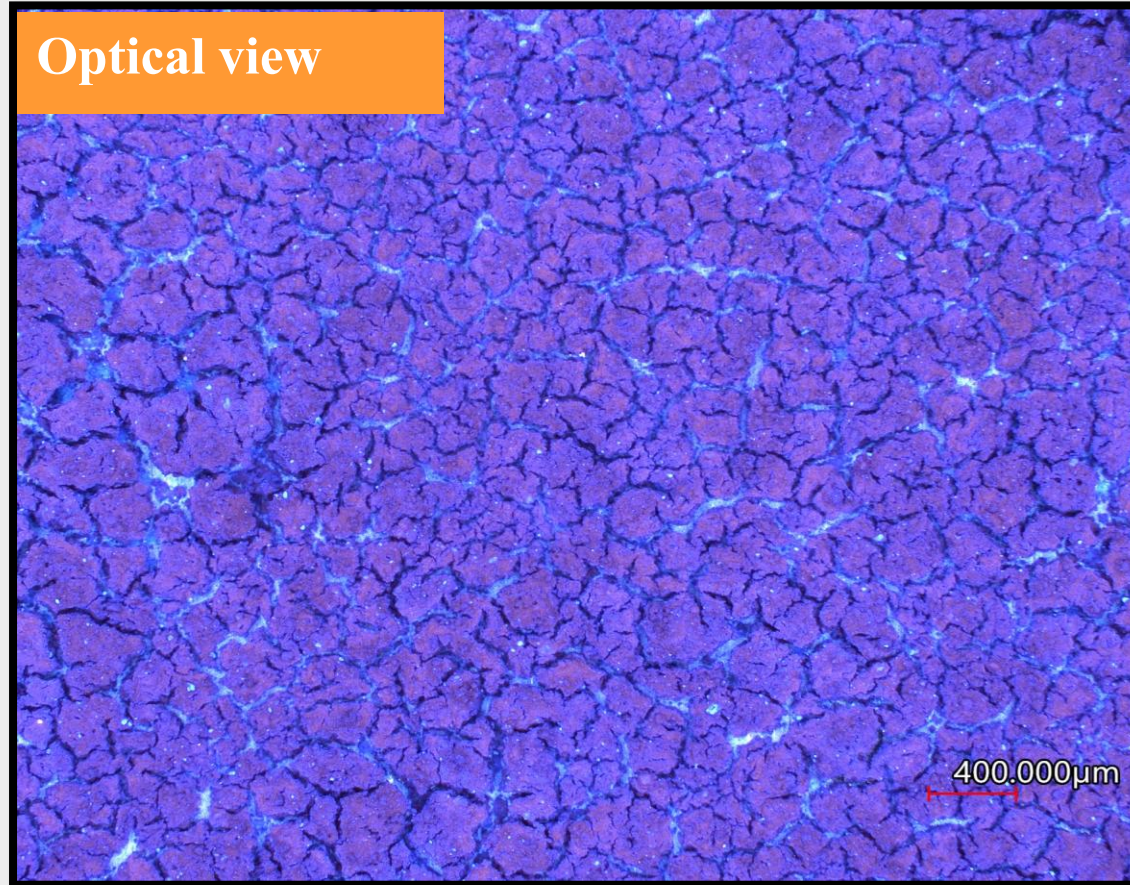
Frequency Sweep



Selected resonant frequencies are swept from lowest to highest (± 200 Hz)

Fabrication Results

Optical view



Best Results*

Step 1: Mix Dye + TMSS+ Ethanol
20cm away

>

Step 2: Stir for 1 hour

>

Step 3: Spray from

Step 4: Multiple sprays

>

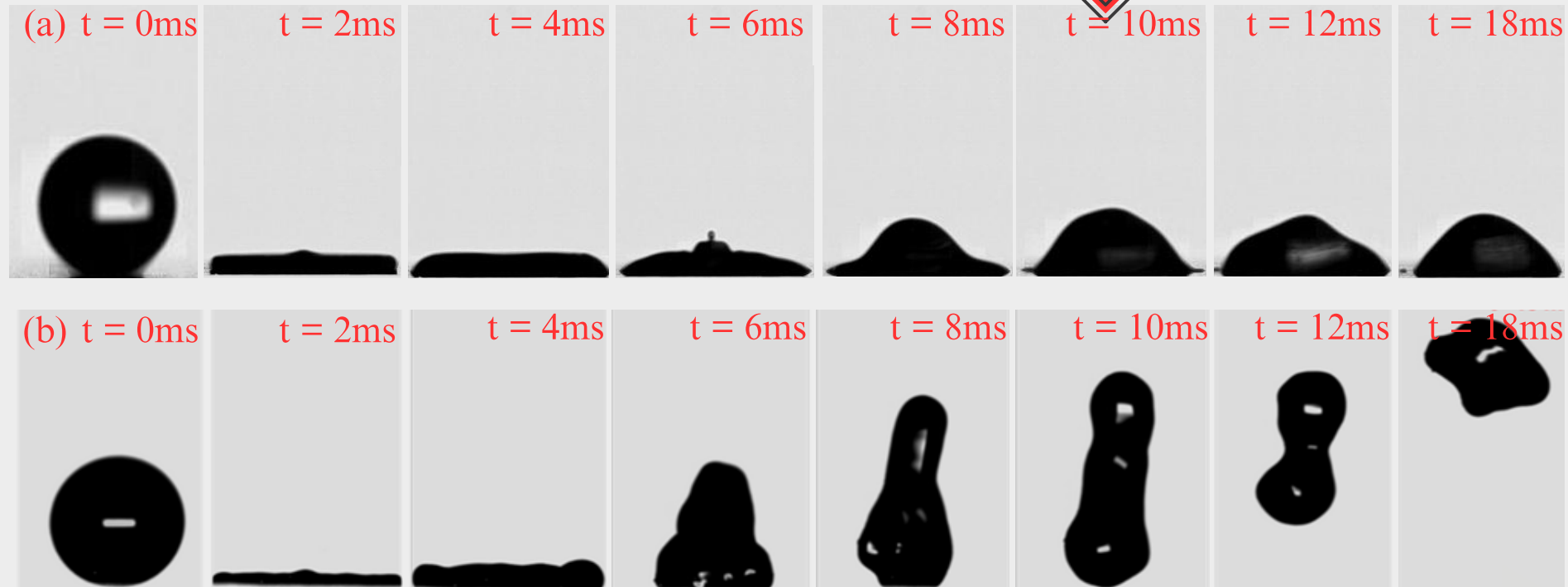
step 5: Dry at room temp

Super Hydrophobic Surface

- $Re: \sim 2175$
Velocity: $\sim 1.11\text{m/s}$
 $We: \sim 32.55$

- (a) Smooth AL

- (b) β -type copper sprayed AL :
 $Ra = 18.32 \mu\text{m}$



Change In impact:

(a) **Pinned** to the surface

(b) **Rebounds** and bounces a couple of times before settling on the surface

Re= 645.20

Re= 2178.81

Re= 4703.56

**Rebound +
Sub-droplet ejection**

Rebound

Splash + Rebound



MERCI



CREATE

Center for Research in Experimental
Aero/Hydrodynamic Technology

Acoustics – Flow – Fluid Structure Interaction

Appendix

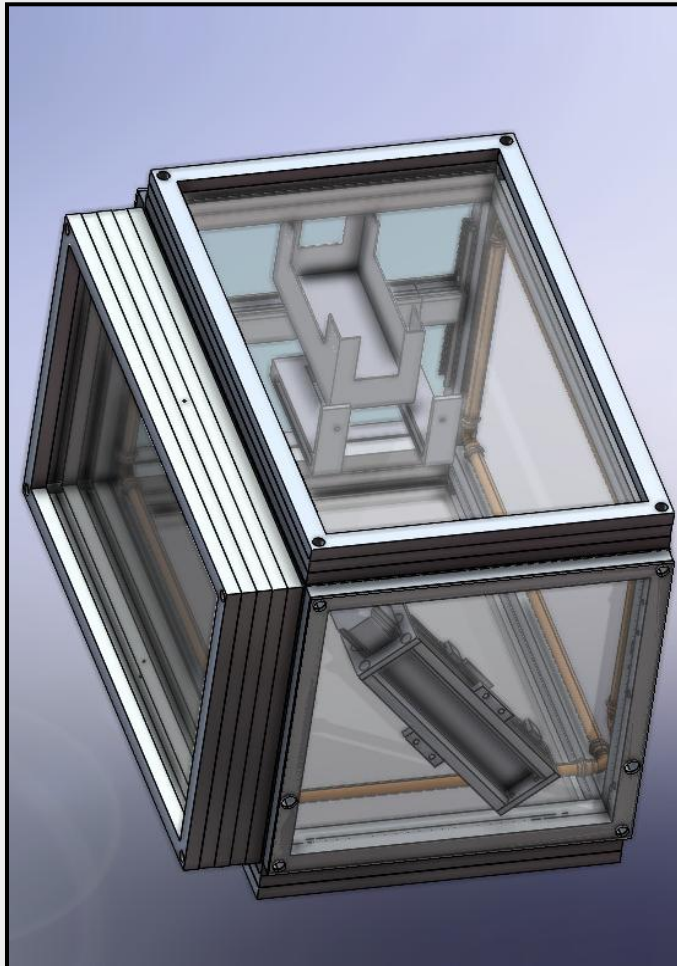
Supercooled Box

Frost control : N2 gas used to reduce humidity

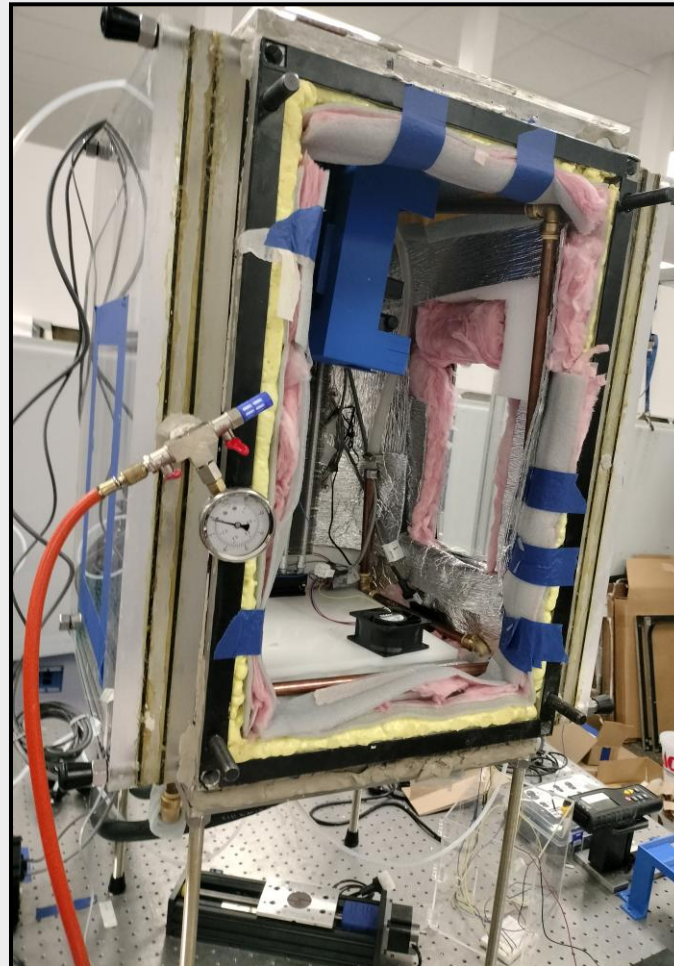
Temperatures (currently): down to -9°C

Cool Agent: Propylene glycol

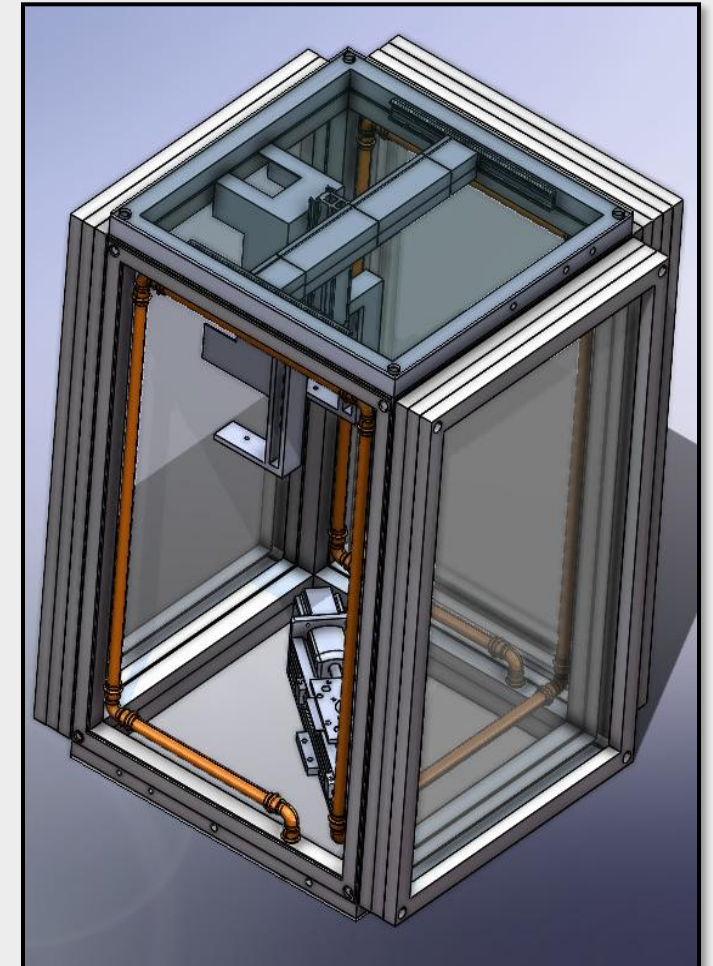
Internal Dim: 400x400x700 mm³



Bottom Isometric view



Front panel opened



Top Isometric view